EXPERIMENT-25:

AIM:

To write a C program to demonstrate that if a plaintext block has a common factor with RSA modulus \(n=pq\), the factor can help recover the prime factors of \(n\) without knowing the private key.

CODE:

```
#include <stdio.h>

int gcd(int a, int b) {
    while (b) {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}

int main() {
    int n, e, m, g;

    printf("Enter RSA modulus n: ");
    scanf("%d", &n);
    printf("Enter public key e: ");
    scanf("%d", &e);
    printf("Enter known plaintext block: ");
```

```
    scanf("%d", &m);

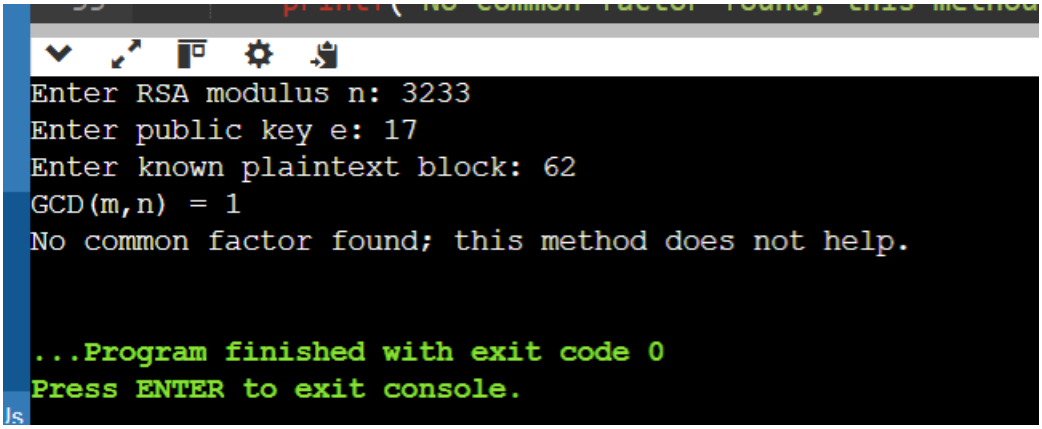
    g = gcd(m, n);

    printf("GCD(m,n) = %d\n", g);

    if (g > 1 && g < n) {
        printf("Factor found: p = %d\n", g);
        printf("Other factor q = %d\n", n / g);
        printf("RSA modulus is factorized!\n");
    } else if (g == n) {
        printf("Plaintext block equals a multiple of n; no proper factor
found.\n");
    } else {
        printf("No common factor found; this method does not help.\n");
    }

    return 0;
}
```

OUTPUT:


Enter RSA modulus n: 3233
Enter public key e: 17
Enter known plaintext block: 62
GCD(m,n) = 1
No common factor found; this method does not help.
...Program finished with exit code 0
Press ENTER to exit console.